

## ULRICHEN, Switzerland

WMO: 067450

Lat: 46.5049N

Lon: 8.3082E

Elev: 1347

StdP: 86.15

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.8      | -18.6 | -23.6                         | 0.5 | -19.2 | -21.7 | 0.6 | -17.4 | 9.2                   | -2.7 | 7.8 | -3.0 | 0.5                   | 70   | 0.649 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.4                   | 26.4            | 15.2 | 24.7 | 14.7 | 23.2 | 14.1 | 16.1                | 24.3 | 15.4 | 22.9 | 14.7 | 21.4 | 3.8                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.6                            | 11.5 | 16.8 | 12.8 | 10.9 | 16.5 | 12.1 | 10.3 | 16.2 | 50.3          | 24.4 | 47.9 | 23.0 | 45.7 | 21.6 | 19.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 9.3               | 7.6  | 6.4 | -24.5                      | 29.0 | 3.5                | 1.6 | -27.0                                              | 30.1 | -29.1      | 31.0 | -31.0      | 31.9 | -33.6      | 33.0 |  |  |
|                   |      |     | DB                         | WB   | DB                 | WB  | DB                                                 | WB   | DB         | WB   | DB         | WB   | DB         | WB   |  |  |
|                   |      |     | -24.7                      | 17.5 | 3.4                | 0.8 | -27.2                                              | 18.0 | -29.2      | 18.5 | -31.1      | 18.9 | -33.6      | 19.5 |  |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 4.7    | -6.9  | -5.1  | -0.2  | 4.7   | 8.6   | 13.2  | 14.9  | 14.3  | 10.8  | 6.9   | 0.2   | -5.4  |
|                                                                   | DBStd       | 8.49   | 4.67  | 4.49  | 3.83  | 3.08  | 3.14  | 3.10  | 2.75  | 2.86  | 2.93  | 3.48  | 4.33  | 4.78  |
|                                                                   | HDD10.0     | 2404   | 524   | 422   | 318   | 161   | 64    | 8     | 2     | 4     | 26    | 103   | 293   | 478   |
|                                                                   | HDD18.3     | 4979   | 783   | 656   | 576   | 410   | 302   | 156   | 109   | 127   | 227   | 354   | 543   | 737   |
|                                                                   | CDD10.0     | 472    | 0     | 0     | 0     | 1     | 21    | 103   | 154   | 137   | 49    | 7     | 0     | 0     |
|                                                                   | CDD18.3     | 5      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 2     | 0     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 318    | 0     | 0     | 0     | 0     | 2     | 83    | 123   | 103   | 8     | 0     | 0     | 0     |
|                                                                   | CDH26.7     | 38     | 0     | 0     | 0     | 0     | 0     | 11    | 12    | 16    | 0     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 1.7    | 1.1   | 1.4   | 1.8   | 2.3   | 2.4   | 2.3   | 2.1   | 1.9   | 1.8   | 1.5   | 1.2   | 0.9   |
| Precipitation                                                     | PrecAvg     | 1579   | 98    | 104   | 103   | 125   | 161   | 154   | 140   | 154   | 138   | 127   | 160   | 115   |
|                                                                   | PrecMax     | 2042   | 241   | 294   | 204   | 232   | 358   | 259   | 255   | 274   | 338   | 302   | 494   | 291   |
|                                                                   | PrecMin     | 1106   | 24    | 20    | 19    | 20    | 45    | 60    | 75    | 48    | 41    | 15    | 31    | 12    |
|                                                                   | PrecStd     | 208    | 58    | 63    | 53    | 57    | 80    | 49    | 46    | 48    | 70    | 68    | 101   | 61    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 4.2   | 7.3   | 12.7  | 18.6  | 23.2  | 27.9  | 28.1  | 28.7  | 24.4  | 19.9  | 13.6  | 5.7   |
|                                                                   |             | MCWB   | 1.6   | 2.0   | 4.7   | 8.8   | 12.8  | 15.4  | 16.2  | 15.5  | 13.5  | 11.4  | 6.7   | 2.1   |
|                                                                   | 2%          | DB     | 2.3   | 5.1   | 10.6  | 16.3  | 20.4  | 25.7  | 26.2  | 26.0  | 22.1  | 17.7  | 10.6  | 3.3   |
|                                                                   |             | MCWB   | 0.8   | 1.1   | 4.1   | 7.9   | 11.3  | 15.0  | 15.2  | 15.4  | 12.7  | 10.0  | 5.6   | 1.1   |
|                                                                   | 5%          | DB     | 1.1   | 3.1   | 8.6   | 14.1  | 18.2  | 23.7  | 24.6  | 24.0  | 20.2  | 15.7  | 8.3   | 1.8   |
|                                                                   |             | MCWB   | 0.2   | 0.1   | 3.0   | 6.7   | 10.4  | 14.3  | 14.7  | 14.7  | 12.1  | 9.3   | 4.7   | 0.7   |
|                                                                   | 10%         | DB     | 0.2   | 1.4   | 6.5   | 11.9  | 16.1  | 21.6  | 23.0  | 22.1  | 18.2  | 13.7  | 6.0   | 0.8   |
|                                                                   |             | MCWB   | -0.6  | -0.3  | 2.1   | 5.6   | 9.2   | 13.4  | 14.1  | 14.1  | 11.2  | 8.3   | 4.1   | 0.1   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 2.5   | 3.0   | 5.6   | 9.5   | 13.2  | 16.7  | 17.1  | 17.0  | 14.7  | 12.4  | 8.3   | 3.1   |
|                                                                   |             | MCDB   | 3.1   | 5.8   | 11.0  | 17.4  | 21.5  | 25.6  | 25.6  | 25.3  | 21.2  | 18.0  | 11.0  | 4.8   |
|                                                                   | 2%          | WB     | 1.2   | 1.6   | 4.5   | 8.2   | 12.0  | 15.7  | 16.2  | 16.0  | 13.7  | 11.1  | 6.5   | 1.7   |
|                                                                   |             | MCDB   | 1.8   | 4.0   | 9.4   | 15.4  | 18.7  | 24.1  | 24.3  | 23.7  | 19.8  | 15.4  | 9.3   | 2.6   |
|                                                                   | 5%          | WB     | 0.3   | 0.7   | 3.5   | 7.2   | 10.9  | 14.7  | 15.4  | 15.2  | 12.8  | 10.1  | 5.3   | 1.0   |
|                                                                   |             | MCDB   | 0.9   | 2.5   | 7.7   | 13.2  | 16.8  | 22.4  | 22.8  | 22.1  | 18.6  | 13.9  | 7.8   | 1.6   |
|                                                                   | 10%         | WB     | -0.4  | -0.1  | 2.5   | 5.9   | 9.8   | 13.8  | 14.6  | 14.5  | 11.9  | 9.2   | 3.9   | 0.2   |
|                                                                   |             | MCDB   | 0.3   | 1.3   | 5.7   | 11.0  | 15.0  | 20.5  | 21.2  | 20.6  | 16.8  | 12.6  | 6.0   | 0.9   |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.5  | 12.2  | 11.7  | 11.3  | 11.5  | 13.3  | 13.4  | 13.0  | 12.8  | 12.1  | 9.2   | 9.3   |
|                                                                   |             | MCDBR  | 9.1   | 14.1  | 15.4  | 16.2  | 16.3  | 18.0  | 17.4  | 17.8  | 17.3  | 15.9  | 12.8  | 9.2   |
|                                                                   | 5% WB       | MCWBR  | 7.6   | 10.3  | 9.2   | 8.3   | 8.0   | 8.2   | 7.4   | 7.9   | 8.5   | 8.9   | 8.0   | 7.0   |
|                                                                   |             | MCDBR  | 7.9   | 11.6  | 13.0  | 14.9  | 14.4  | 16.6  | 15.7  | 15.4  | 14.8  | 13.1  | 11.3  | 7.3   |
|                                                                   |             | MCWBR  | 6.7   | 8.7   | 7.8   | 7.9   | 7.3   | 7.7   | 6.9   | 7.2   | 7.4   | 7.6   | 7.4   | 5.8   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.252 | 0.281 | 0.331 | 0.382 | 0.403 | 0.410 | 0.407 | 0.395 | 0.366 | 0.334 | 0.288 | 0.250 |
|                                                                   | taud        |        | 2.290 | 2.213 | 2.129 | 2.099 | 2.159 | 2.197 | 2.207 | 2.257 | 2.322 | 2.381 | 2.389 | 2.326 |
|                                                                   | Ebn at Noon |        | 852   | 889   | 886   | 869   | 863   | 856   | 853   | 849   | 843   | 819   | 801   | 814   |
|                                                                   | Edh at Noon |        | 76    | 102   | 130   | 148   | 145   | 141   | 137   | 126   | 107   | 87    | 70    | 65    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 1.57  | 2.52  | 3.72  | 4.45  | 4.81  | 5.67  | 5.65  | 4.91  | 3.83  | 2.54  | 1.55  | 1.30  |
|                                                                   | RadStd      |        | 0.19  | 0.30  | 0.54  | 0.71  | 0.47  | 0.51  | 0.49  | 0.44  | 0.37  | 0.29  | 0.23  | 0.17  |

## Historical Trends

|                               | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                               |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                               |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only             | N/A   | +3.11   | +3.70  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (119 neighbors) | N/A   | N/A     | N/A    | +0.73   | +0.42 | +0.32 | N/A         | N/A     | +58     | +16     |

Nomenclature: See separate page